ICS Risk Assessment

1 Risk Matrix

Table 1: ICS Threat Risk Assessment

Threat	Likelihood	Impact (Safety / Production / Education)	Risk Score	Priority
Engineering PC malware infection	High (3)	Critical (4): full compromise of control layer, persistent attacker access	12	Critical
PLC logic manipulation	Medium (2)	Critical (4): unsafe actuator behaviour, production shutdown, learning disruption	8	High
Unauthorized engineering access (insider misuse)	Medium (2)	Critical (4): silent logic changes, safety bypass possible	8	High
SCADA/HMI denial of service	Medium (2)	High (3): loss of operator visibility, production interruption	6	Medium
Network flooding / traffic storm	Medium (2)	High (3): loss of PLC communication, system downtime	6	Medium
MES production manipulation	Medium (2)	High (3): incorrect production runs, financial + operational impact	6	Medium
PLC configuration/data exfiltration	Medium (2)	High (3): loss of intellectual property + enables future attacks	6	Medium
Rogue device in network	Medium (2)	Medium (2): reconnaissance, lateral movement	4	Medium
Webshop/API data manipulation	Medium (2)	Medium (2): incorrect orders, indirect production issues	4	Medium
PLC command spoofing	Low (1)	High (3): incorrect control actions, potential physical instability	3	Low
Session hijacking (SCADA)	Low (1)	High (3): stealth control manipulation, loss of trust in system	3	Low